

Making Visible and Keeping Concealed: Architectures of Melen Water Supply Project

Elif Bek

Abstract

This research asks what kinds of architecture a territorial water infrastructure produces. It examines this question through the Melen Water Supply Project, a system that supplies drinking water to Istanbul and that operates while the reservoir giving the project its name has not yet been impounded. Only a small part of the interventions carried out in the basin is accessible as information. This research therefore focuses on the question of visibility: how does architecture make an otherwise largely invisible water infrastructure visible, legible, and publicly imaginable, and, conversely, how does it conceal, obscure, or restrict access to it?

The architectural structures and spaces that constitute the Melen system will be examined with consideration of visibility not simply as a visual condition but in relation to legibility, knowledge, access, representation, and publicness. These structures that define the relationship with water also define how interaction with water gave shape to the built environment across supply, sanitation, and flood mitigation alike (Murphy, 2024).

Each register within the Melen Water Supply Project can be approached through a different set of elements that contain the issue of visibility. Legibility can be read in the structures that are passed without being recognised for what they are, the intake regulator and the pumping station among them. It can also be read in the treatment plants, which are recognisable as a building type but not as part of a project that supplies another city. Knowledge can be traced in what is documented and released, since much of the architecture appears publicly only through a protection plan and its annexed map, including the interception collectors that run buried alongside the watercourses and the dispersed network of wastewater facilities. Access can be observed in the fence, the gate, the service road, and the distance the regulations prescribe between recreational use and the water intake. Representation can be located in the drawing, the photograph, and the caption through which this dual system of supply and sanitation can be defined in its entirety. Publicness, understood here as visibility in relation to public use, can be discussed in elements such as the filtration strips, bands of land along the watercourses that the administration expropriates for İSKİ, grades, and plants so that vegetation filters surface runoff before it reaches the water.

The questions of how architecture mediates between the technical infrastructure of water and what becomes visible, to whom, and through what architectural forms and spatial arrangements will be posed. The Melen Water Supply Project comprises structures that are approached and used, as well as structures that are passed without notice at the "infra" as the spaces of water that become visible and invisible intermittently within the basin. By setting the visible and the invisible of the project's architecture, and the relationalities between them will illustrate the stance on the interaction of water within the city.

*Murphy, C. E. (2024). Architectures of Water in Early Modernity. Massachusetts Institute of Technology, Department of Architecture.
<https://architecture.mit.edu/classes/special-subject-history-theory-criticism-architecture-art-architectures-water-early>

Literature Review

The literature review conducted within this research was built on core readings that address the visibility of infrastructure and on the question of whether the research is positioned within visual culture or critical geography, taking visibility in relation to legibility, knowledge, access, representation, and publicness as its framework. The TU Delft Repository, DSpace@MIT and MIT Primo, HOLLIS, JSTOR, Google Scholar, Crossref, OpenMETU and Academia.edu were searched with a focus on architecture-focused sources in both fields and on sources that address the question of visibility, and each reference was verified through Crossref, publisher pages, repository records, or the full text.

Of the thirty sources included in this review, twenty-one address the question of visibility. The remaining nine comprise four architecture-focused sources in visual culture, one architecture-focused source in critical geography, one other source with a strong architectural approach, two sources with a weak architectural approach and one master's thesis.

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	POSITIONING	VERIFIED LINK
Larkin, B. (2013). The politics and poetics of infrastructure. <i>Annual Review of Anthropology</i> , 42, 327–343. https://doi.org/10.1146/annurev-anthro-092412-155522	In the full text (p. 336), Larkin questions the repetition of Star's (1999) assertion that infrastructures are "by definition invisible", stating that invisibility "is only one and at the extreme edge of a range of visibilities that move from unseen to grand spectacles and everything in between", and proposes "to examine how (in)visibility is mobilized and why". The question of visibility in this research can be defined with reference to this range.	FIELD Anthropology ARCHITECTURAL APPROACH Weak AXIS Closer to visual culture (open to critical geography through biopolitics and technopolitics) BASIS (VERIFIED WORDING) Abstract: "the aesthetic and the sensorial, desire and promise". p. 336: invisibility as the extreme edge of "a range of visibilities that move from unseen to grand spectacles"	https://doi.org/10.1146/annurev-anthro-092412-155522 Verified through: Crossref
Star, S. L., & Ruhleder, K. (1996). Steps toward an ecology of infrastructure: Design and access for large information spaces. <i>Information Systems Research</i> , 7 (1), 111–134. https://doi.org/10.1287/isre.7.1.111	On p. 113, "Becomes visible upon breakdown" is listed among the dimensions of infrastructure: "The normally invisible quality of working infrastructure becomes visible when it breaks." The same page holds that "infrastructure is a fundamentally relational concept" and takes the city water system as its example: the cook considers it "a piece of working infrastructure integral to making dinner; for the city planner, it becomes a variable in a complex equation."	FIELD Science and technology studies ARCHITECTURAL APPROACH Weak AXIS Outside both axes, shared ground (its relational definition sits closer to critical geography) BASIS (VERIFIED WORDING) p. 113: "Becomes visible upon breakdown"; "a fundamentally relational concept"; the city water system example (the cook and the city planner)	https://doi.org/10.1287/isre.7.1.111 Verified through: JSTOR full text (METU access)
Foucault, M. (1977). <i>Discipline and punish: The birth of the prison</i> (A. Sheridan, Trans.). Pantheon Books. (Original work published 1975)	On p. 200, Bentham's Panopticon is described as "the architectural figure of this composition", followed by the sentence "Visibility is a trap." On p. 201, the major effect of the Panopticon is "to induce in the inmate a state of conscious and permanent visibility that assures the automatic functioning of power." The relation between architecture, visibility and power in this research can be traced to these pages.	FIELD Philosophy ARCHITECTURAL APPROACH Strong AXIS Visual culture (the text's own field is disciplinary power) BASIS (VERIFIED WORDING) p. 200: the Panopticon as "the architectural figure of this composition"; "Visibility is a trap." p. 201: "conscious and permanent visibility"	https://archive.org/details/disciplinepunish000fouc_j5k6/page/200/mode/2up?q=%22visibility+is+a+trap%22 Verified through: Internet Archive search inside (Pantheon 1977)
Star, S. L. (1999). The ethnography of infrastructure. <i>American Behavioral Scientist</i> , 43(3), 377–391. https://doi.org/10.1177/00027649921955326	The publisher's abstract describes infrastructure as "both relational and ecological" and lists, among the ways of studying it, "understanding the paradoxes of infrastructure as both transparent and opaque" and "including invisible work in the ecological analysis". Larkin (2013, p. 336) identifies this article as the source of the assertion that infrastructures are "by definition invisible".	FIELD Science and technology studies ARCHITECTURAL APPROACH Weak AXIS Outside both axes, the conceptual ground they share BASIS (VERIFIED WORDING) Abstract: "both relational and ecological"; "the paradoxes of infrastructure as both transparent and opaque"	https://doi.org/10.1177/00027649921955326 Verified through: Crossref

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	POSITIONING	VERIFIED LINK
Deleuze, G. (1988). <i>Foucault</i> (S. Hand, Trans.). University of Minnesota Press. (Original work published 1986)	On p. 32, Deleuze writes that the prison "constitutes a visibility" and is "a system of light before being a figure of stone", and introduces Foucault's distinction between "the visible and the articulable". The register of knowledge in this research can be discussed through this distinction. Architectural approach: weak.		https://archive.org/details/foucault0000dele/page/32/mode/2up?q=%22the+visible+and+the+articulable%22 Verified through: Internet Archive search inside (University of Minnesota Press 1988)
Atas, Z., Atmaca, Y., & Tasguzen Polat, Z. (2026). Historic water infrastructure as an urban-spatial system in the old city of Mardin: A multi-scalar spatial-decoding approach. <i>Urban Science</i> , 10(8), 428. https://doi.org/10.3390/urbansci10080428	According to the publisher's abstract, in the old city of Mardin "the transition to centralized modern supply" transformed "the visibility, accessibility, use, and situated knowledge of water", and the study reconstructs the relationships "between partially concealed infrastructure and visible urban form". Architectural approach: strong.		https://doi.org/10.3390/urbansci10080428 Verified through: Crossref, MIT Primo and HOLLIS (Harvard)
Biersteker, E., Koppenjan, J., & van Marrewijk, A. (2021). Translating the invisible: Governing underground utilities in the Amsterdam airport Schiphol terminal project. <i>International Journal of Project Management</i> , 39(6), 581–593. https://doi.org/10.1016/j.ijproman.2021.04.003	According to the abstract, the article draws on "translation theory, which is essentially about the reinterpretation, appropriation, and representation of interests related to materials", and shows that "the translation of the underground by project actors hampered the necessary relocation of utilities". The register of knowledge can be traced in this account of underground utilities. Architectural approach: weak.		https://doi.org/10.1016/j.ijproman.2021.04.003 Verified through: Crossref and TU Delft Repository
Bowker, G. C., & Star, S. L. (1999). <i>Sorting things out: Classification and its consequences</i> . MIT Press. https://doi.org/10.7551/mitpress/6352.001.0001	According to the publisher's abstract, the authors "emphasize the role of invisibility in the process by which classification orders human interaction", examine "how categories are made and kept invisible, and how people can change this invisibility when necessary", and treat systems of classification "as part of the built information environment". The register of knowledge can be traced through this account of classification. Architectural approach: weak.		https://doi.org/10.7551/mitpress/6352.001.0001 Verified through: Crossref
da Cunha, D. (2019). <i>The invention of rivers: Alexander's eye and Ganga's descent</i> . University of Pennsylvania Press.	According to the publisher's description, rivers are treated as "products of human design and a particular way of seeing that has roots stretching as far back as ancient Greek cartography", and the book traces how the Ganges "has been pictured, mapped, surveyed, explored, and measured across the millennia"; its first chapter is titled "Alexander's Eye and Ganga's Descent". The register of representation can be discussed through this way of seeing. Architectural approach: strong.		https://www.pennpress.org/9780812249996/the-invention-of-rivers/ Verified through: Penn Press publisher page and review in Planning Perspectives

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	POSITIONING	VERIFIED LINK
Easterling, K. (2014). <i>Extrastatecraft: The power of infrastructure space</i> . Verso.	According to the publisher's description, "infrastructure sets the invisible rules that govern the spaces of our everyday lives", and the book reveals governmental and corporate forces "buried within the concrete and fiber-optics of our modern habitat". The register of knowledge can be discussed in relation to these "invisible rules" of infrastructure space. Architectural approach: strong.		https://www.versobooks.com/products/30-extrastatecraft Verified through: Verso publisher page
Gandy, M. (2014). <i>The fabric of space: Water, modernity, and the urban imagination</i> . MIT Press. https://doi.org/10.7551/mitpress/8313.001.0001	According to the publisher's abstract, water lies "at the intersection of landscape and infrastructure, crossing between visible and invisible domains of urban space"; the book begins with "the Parisian sewers of the nineteenth century, captured in the photographs of Nadar" and addresses Berlin's "protection of public access to lakes for swimming". The registers of representation and access can be read in these cases. Architectural approach: strong.		https://doi.org/10.7551/mitpress/8313.001.0001 Verified through: Crossref
Hutton, J. (2019). <i>Reciprocal landscapes: Stories of material movements</i> . Routledge. https://doi.org/10.4324/9781315737102	The publisher's description opens with the question: "How are the far-away, invisible landscapes where materials come from related to the highly visible, urban landscapes where those same materials are installed?" The book traces five construction materials "from seminal public landscapes in New York City, back to where they came from". Architectural approach: strong.		https://doi.org/10.4324/9781315737102 Verified through: Crossref
Kaika, M. (2006). Dams as symbols of modernization: The urbanization of nature between geographical imagination and materiality. <i>Annals of the Association of American Geographers</i> , 96(2), 276–301. https://doi.org/10.1111/j.1467-8306.2006.00478.x	According to the publisher's abstract, the article analyses "the iconography and symbolism of dam constructions" through the Marathon dam, which "became an iconic marker of Athens's modernization" and was "veneered with the same marble as that used for the Parthenon", read as an effort "to reconcile iconographically the two prevailing geographical imaginations" of the city. The register of representation can be located in this reading of the dam as architecture. Architectural approach: strong.		https://doi.org/10.1111/j.1467-8306.2006.00478.x Verified through: Crossref and HOLLIS (Harvard)

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	POSITIONING	VERIFIED LINK
Kaika, M., & Swyngedouw, E. (2000). Fetishizing the modern city: The phantasmagoria of urban technological networks. <i>International Journal of Urban and Regional Research</i> , 24(1), 120–138. https://doi.org/10.1111/1468-2427.00239	According to the publisher's abstract, "Burying the flow of water via subterranean and often distant pinpointed technological mediations (dams, purification plants, pumping stations) facilitates and contributes to masking the social relations through which the metabolic urbanization of water takes place"; in early modernity such structures "were celebrated as glorious icons, carefully designed, ornamented and prominently located in the city", whereas in high modernity they "began to disappear from the cityscape" and "from the urban imagination". Architectural approach: strong.		https://doi.org/10.1111/1468-2427.00239 Verified through: Crossref
Kılıç, B. Ş. (2020). <i>Questioning the 'modern water' of İstanbul: Proposing a new approach to design urban water infrastructure of the city</i> [Master's thesis, Delft University of Technology]. TU Delft Repository. https://resolver.tudelft.nl/uuid:4cc2af52-4a59-451b-9d48-a3830f3d6b17	According to the abstract of this master's thesis, the "'Modern Water' understanding of today sees water as a service to be provided and makes it invisible to users", causing "lack of awareness, responsibility and loss of a socio-cultural value". The thesis concerns İstanbul, the city supplied by the Melen Water Supply Project. Architectural approach: strong.		https://resolver.tudelft.nl/uuid:4cc2af52-4a59-451b-9d48-a3830f3d6b17 Verified through: TU Delft Repository record
Mitchell, W. J. T. (2002). Showing seeing: A critique of visual culture. <i>Journal of Visual Culture</i> , 1(2), 165–181. https://doi.org/10.1177/147041290200100202	According to the publisher's abstract, the essay maps out "visual studies as an emergent academic formation", questions "the overturning of 'scopic regimes'" and concludes with an exercise the author calls "showing seeing". The source can frame the positioning of this research in relation to visual culture. Architectural approach: weak.		https://doi.org/10.1177/147041290200100202 Verified through: Crossref
Mattern, S. (2013, July). Infrastructural tourism. <i>Places Journal</i> . https://doi.org/10.22269/130701	The essay, open access in Places Journal, discusses tours, audio guides and artistic projects that aim to "make visible the invisible", including a project that highlights what remains "at the periphery of visibility," what seems "illegible or even un-mappable". The registers of publicness and legibility can be discussed through these practices of infrastructural tourism. Architectural approach: strong.		https://doi.org/10.22269/130701 Verified through: Crossref
Atkinson, T. (2025). <i>Monumental designs: Infrastructure and the culture of the Tennessee Valley Authority</i> . University Press of Mississippi. https://doi.org/10.2307/jj.25465122	The first chapter, "A Dam Site More Modern: Developing TVA Monumentality in Visual Culture" (pp. 24-56), situates the infrastructure of the Tennessee Valley Authority within visual culture. The register of representation can be located in this material. Architectural approach: strong.		https://doi.org/10.2307/jj.25465122 Verified through: JSTOR search result and Crossref

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	POSITIONING	VERIFIED LINK
Colven, E. (2020). Subterranean infrastructures in a sinking city: The politics of visibility in Jakarta. <i>Critical Asian Studies</i> , 52(3), 311–331. https://doi.org/10.1080/14672715.2020.1793210	According to the publisher's abstract, the article explores "the politics of infrastructure in Jakarta through the lens of in/visibility", notes that infrastructure studies has tended to categorise infrastructure "as either hyper-visible by design, or invisible until breakdown", and argues that "spectacular, visible infrastructures generate public and political attention, while below ground, hidden and invisible infrastructures are overlooked". The register of publicness can be discussed through this "politics of visibility". Architectural approach: weak.		https://doi.org/10.1080/14672715.2020.1793210 Verified through: Crossref
Kambic, K. (2015). Urban water visibility in Los Angeles: Legibility and access for all. In S. Buechler & A.-M. Hanson (Eds.), <i>A political ecology of women, water, and global environmental change</i> (pp. 58–76). Routledge. https://doi.org/10.4324/9781315796208-5	According to the chapter abstract, Kambic examines "three responses to inaccessible water infrastructure" in Los Angeles, namely "accepting the situation, incremental infrastructure, and the Cyborg City", under the title of urban water visibility, legibility and access. The registers of legibility and access can be read in these responses. Architectural approach: weak.		https://doi.org/10.4324/9781315796208-5 Verified through: Crossref
Sarıkaya Işık, F. (2019). <i>Byzantine heritage depicted: The Aqueduct of Valens in the historical topography of Istanbul</i> [Master's thesis, Middle East Technical University]. OpenMETU. https://open.metu.edu.tr/handle/11511/45494	The title presents the Aqueduct of Valens as "Byzantine heritage depicted", and the abstract states that the aqueduct "maintained its critical operation for supplying water and, at the same time protected its urban presence in the urban topography". The register of representation can be located in the depictions of a water structure of Istanbul. Architectural approach: strong.		https://open.metu.edu.tr/handle/11511/45494 Verified through: OpenMETU record and METU History of Architecture MA theses list

Visual culture: architecture-focused sources

4 sources

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Ghosn, R., & Jazairy, E. H. (2018). <i>Geostories: Another architecture for the environment</i> . Actar.	According to the publisher's description, the architectural projects of the book become "a medium to synthesize different forms and scales of knowledge on technological externalities", among them water shortage, and revisit "devices of wonder that assemble publics around representations of the Earth". The registers of representation and publicness can be discussed through this design research. Architectural approach: strong.	https://www.design-earth.org/publications/geostories/ Verified through: Design Earth publication page and review in The Journal of Architecture (Crossref)
Mathur, A., & da Cunha, D. (2001). <i>Mississippi floods: Designing a shifting landscape</i> . Yale University Press.	The record held for this review is limited to the bibliographic details and a book review; the book addresses the Mississippi floods through "designing a shifting landscape", as its title states. Its relation to the register of representation remains to be established by reading. Architectural approach: strong.	https://api.crossref.org/works/10.3368/lj.21.1.220 Verified through: Book review in Landscape Journal (Waldheim, 2002) and Crossref

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Mathur, A., & da Cunha, D. (2016). Aqueous terrain. <i>Journal of Architectural Education</i> , 70(1), 35–37. https://doi.org/10.1080/10464883.2016.1128262	A short text (pp. 35-37) in the Journal of Architectural Education titled "Aqueous Terrain". The record held for this review is limited to the bibliographic details; its relation to the question of visibility remains to be established by reading. Architectural approach: strong.	https://doi.org/10.1080/10464883.2016.1128262 Verified through: Crossref
Varnelis, K. (Ed.). (2008). <i>The infrastructural city: Networked ecologies in Los Angeles</i> . Actar.	The review of the book in Places Journal describes Los Angeles as depending "upon vast infrastructural systems that are breathtakingly complex, yet vulnerable to disruption, even disaster". The book is an edited volume on the networked ecologies of the city; its chapters remain to be read for the question of visibility. Architectural approach: strong.	https://api.crossref.org/works/10.1111/j.1531-314x.2010.01114.x Verified through: Review in the Journal of Architectural Education (Osman, 2010, Crossref) and review in Places Journal

Critical geography: architecture-focused sources

1 source

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Hein, C. (Ed.). (2020). <i>Adaptive strategies for water heritage: Past, present and future</i> . Springer. https://doi.org/10.1007/978-3-030-00268-8	According to the book description, people have created around water "an extensive tangible network of infrastructure, buildings, and urban form", while today "heritage and water are largely two separate disciplinary and professional fields"; the volume links the two through twenty-one chapters. Architectural approach: strong.	https://doi.org/10.1007/978-3-030-00268-8 Verified through: Crossref

Other sources with a strong architectural approach

1 source

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Murphy, C. E. (2024). <i>Architectures of water in early modernity</i> . MIT Department of Architecture. https://architecture.mit.edu/classes/special-subject-history-theory-criticism-architecture-art-architectures-water-early	A course description of the MIT Department of Architecture, cited in this research; it is not a publication, and the texts on its reading list can be consulted as citable sources. Architectural approach: strong.	https://architecture.mit.edu/classes/special-subject-history-theory-criticism-architecture-art-architectures-water-early Verified through: MIT Architecture page

Sources with a weak architectural approach

2 sources

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Carse, A. (2016). Keyword: Infrastructure. How a humble French engineering term shaped the modern world. In P. Harvey, C. B. Jensen, & A. Morita (Eds.), <i>Infrastructures and social complexity: A companion</i> (pp. 45–57). Routledge. https://doi.org/10.4324/9781315622880-11	As the title states, the chapter traces how "a humble French engineering term shaped the modern world". It can serve the definition of the term infrastructure in this research. Architectural approach: weak.	https://doi.org/10.4324/9781315622880-11 Verified through: Taylor & Francis chapter page

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Graham, S., & Marvin, S. (2001). <i>Splintering urbanism: Networked infrastructures, technological mobilities and the urban condition</i> . Routledge. https://doi.org/10.4324/9780203452202	According to the publisher's description, the book analyses "the complex interactions between infrastructure networks and urban spaces", including "infrastructure and the built environment", under the notion of splintering urbanism. Architectural approach: weak.	https://doi.org/10.4324/9780203452202 Verified through: Crossref

Master's thesis

1 source

SOURCE (APA 7)	RELEVANCE TO THIS RESEARCH	VERIFIED LINK
Demirtaş, F. A. (2000). <i>Artificial nature: Water infrastructure and its experience as natural space</i> [Master's thesis, Massachusetts Institute of Technology]. DSpace@MIT. http://hdl.handle.net/1721.1/65716	According to the abstract of this master's thesis (MIT Department of Architecture, advised by William Porter and Sibel Bozdoğan), the work is about "water infrastructure and its experience as urban and natural space", analysing "water dams and reservoirs that are more than utilitarian structures" through the concept of "ARTIFICIAL NATURE", with sites "from Thrace to Central and Southeastern Anatolia in Turkey from 1920-2000". Architectural approach: strong.	http://hdl.handle.net/1721.1/65716 Verified through: DSpace@MIT record